

Project Documentation Guidelines

Item	DEPI Industry
Project Planning & Management	2 Oct 2026
Literature Review	9 Oct 2026
Requirements Gathering	9 Oct 2026
System Analysis & Design	9 Oct 2026
Implementation (Source Code & Execution)	16 Oct 2026
Final Presentation & Testing& Reports	16 Oct 2026

All documents should be uploaded to GitHub.

1. Project Planning & Management

- **Project Proposal** – Overview of the project, objectives, and scope.
- **Project Plan** – Timeline (Gantt chart), milestones, deliverables, and resource allocation.
- **Task Assignment & Roles** – Defined responsibilities for team members.
- **Risk Assessment & Mitigation Plan** – Identifying risks and solutions.
- **KPIs (Key Performance Indicators)** – Metrics for project success (e.g., response time, system uptime, user adoption rate).

2. Literature Review

- **Feedback & Evaluation** – Lecturer's assessment of the project.
- **Suggested Improvements** – Areas where the project can be enhanced.

- **Final Grading Criteria** – Breakdown of marks based on documentation, implementation, testing, and presentation.

3. Requirements Gathering

- **Stakeholder Analysis** – Identifying key stakeholders and their needs.
 - **User Stories & Use Cases** – Scenarios illustrating how users interact with the system.
 - **Functional Requirements** – List of features and functionalities.
 - **Non-functional Requirements** – Performance, security, usability, and reliability criteria.
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4. System Analysis & Design

1. Problem Statement & Objectives – Define the problem being solved and project goals.

- Use Case Diagram & Descriptions – Identify system actors and interactions.
- Functional & Non-Functional Requirements – Clearly state system capabilities and constraints.
- Software Architecture – High-level design outlining system components, interactions, and architecture style (e.g., MVC, Microservices).

2. Database Design & Data Modeling

- ER Diagram (Entity-Relationship Diagram) – A well-defined ERD showcasing database structure and relationships.
- Logical & Physical Schema – Tables, attributes, keys, and normalization considerations.

3. Data Flow & System Behavior

- DFD (Data Flow Diagram) – Context-level and detailed levels showing how data moves through the system.
- Sequence Diagrams – Process flow representation of key interactions between components.
- Activity Diagram – Visualizing the workflow of processes or user actions within the system.
- State Diagram – Represents different states of an object and how it transitions between them.
- Class Diagram – Defines the structure of the system by showing classes, attributes, methods, and relationships.

4. UI/UX Design & Prototyping

- Wireframes & Mockups – Screens and visual representations of the user interface.
- UI/UX Guidelines – Design principles, color schemes, typography, and accessibility considerations.

5. System Deployment & Integration

- Technology Stack – Backend, frontend, and database technologies.
- Deployment Diagram – Describes how software components are distributed across hardware.
- Component Diagram – Shows high-level system components and their dependencies.

6. Additional Deliverables (if applicable)

- API Documentation – If the system includes APIs, provide documentation for endpoints and usage.
- Testing & Validation – Unit tests, integration tests, and user acceptance testing plan.
- Deployment Strategy – Hosting environment, deployment pipelines, and scaling considerations.

5. Implementation (Source Code & Execution)

1. Source Code

- Structured & Well-Commented Code – Clean, maintainable, and properly documented code following best practices.
- Coding Standards & Naming Conventions – Consistent formatting, meaningful variable names, and adherence to industry standards.
- Modular Code & Reusability – Organized into reusable components, functions, and classes.
- Security & Error Handling – Secure coding practices, validation checks, and proper exception handling.

2. Version Control & Collaboration

- Version Control Repository – Hosted on GitHub, GitLab, or Bitbucket with a public/private repository link.
- Branching Strategy – Clear workflow (e.g., GitFlow, Feature Branching) for managing code updates.
- Commit History & Documentation – Meaningful commit messages and detailed pull request descriptions.
- CI/CD Integration (if applicable) – Automated builds, testing, and deployment pipelines.

3. Deployment & Execution

- **README File – Includes:**
- **Installation steps**
- **System requirements (hardware/software dependencies)**

- **Configuration instructions**
 - **Execution guide (running the project locally or accessing a deployed version)**
 - **API documentation (if applicable)**
 - **Executable Files & Deployment Link –**
 - **Compiled software or packaged application (e.g., .exe, .jar, .apk).**
 - **Deployed web/mobile app**
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6. Testing & Quality Assurance

- **Test Cases & Test Plan** – Document detailing test scenarios and expected outcomes.
 - **Automated Testing (if applicable)** – Any automated test scripts used.
 - **Bug Reports** – Issues identified and resolutions.
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7. Final Presentation & Reports

- **User Manual** – Instructions for end users.
- **Technical Documentation** – System architecture, database schema, API documentation.
- **Project Presentation (PPT/PDF)** – Summary of the project, challenges, solutions, and outcomes.
- **Video Demonstration (Optional)** – Short demo showcasing the project's functionality.